

Econ 695 - Midterm 1

October 7, 2025

Date and Time: 8:00-9:15 AM, October 8th, 2025

Instructions: Write legibly, and explain as carefully as you can what you are doing so we can award partial credit!

Question 1 (25 points; 15 minutes)

Consider the distribution of individual incomes, y_i in a population that is divided into two groups, indicated by the random variable $D_i \in \{0, 1\}$. The population regression of interest is

$$y_i = \beta_0 + \beta_1 D_i + u_i \quad (1)$$

which satisfies $E[u_i] = E[u_i D_i] = 0$. Suppose that $Pr(D_i = 1) = p$.

(a) (10 points) Find $E[y_i]$ - the population mean of y_i as a function of β_0, β_1 .

(b) (15 points) Suppose $D_i = 1$ if person i is female and 0 otherwise. And we also have $X_i = 1$ if i has at least a college degree, and 0 otherwise. What is the conditional expectation function $E[y_i | D_i, X_i]$? Propose a population regression that coincides with the CEF and explain. *Hint: carefully define the regressors based on binary D_i and binary X_i .*

Question 2 (35 points; 25 minutes)

Consider the following long and short regressions in the population:

$$\begin{aligned} \text{(long)} \quad y_i &= \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i \\ \text{(short)} \quad y_i &= \alpha_0 + \alpha_1 x_{1i} + \alpha_2 x_{2i} + v_i \end{aligned} \quad (2)$$

in which $(\beta_0, \beta_1, \beta_2, \beta_3)$ in the long regression is the unique solution that solves $\min_b E[(y_i - b_0 - b_1 x_{1i} - b_2 x_{2i} - b_3 x_{3i})^2]$, and $(\alpha_0, \alpha_1, \alpha_2)$ in the short regression is the unique solution that solves $\min_a E[(y_i - a_0 - a_1 x_{1i} - a_2 x_{2i})^2]$.

(a) (10 points) Derive a formula relating α_2 and β_2 on x_{2i} .

(b) (10 points) Suppose that $cov(x_{3i}, x_{1i}) = 0$ and $cov(x_{3i}, x_{2i}) = 0$. Does $\alpha_2 = \beta_2$? Explain.

(c) (15 points) Suppose that

$$\text{cov}(x_{2i}, x_{3i}) = 0, \text{cov}(x_{1i}, x_{3i}) \neq 0, \text{ and } \text{cov}(x_{1i}, x_{2i}) \neq 0$$

Does $\alpha_2 = \beta_2$? Explain.

Question 3 (40 points; 30 minutes)

In Lecture 3b we examined the role of education in driving the wage gap between native and immigrants. We considered four OLS regressions:

$$(1) \hat{y}_i = \hat{\pi}_0 + \hat{\pi}_1 \text{Imm}_i$$

$$(2) \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{Educ}_i + \hat{\gamma} \text{Imm}_i$$

$$(3) \hat{y}_i = \hat{\beta}_0^a + \hat{\beta}_1^a \text{Educ}_i \text{ among natives}$$

$$(4) \hat{y}_i = \hat{\beta}_0^b + \hat{\beta}_1^b \text{Educ}_i \text{ among immigrants}$$

$$\Rightarrow \text{Educ}_i^a = \frac{\bar{y}^a - \hat{\beta}_0^a}{\hat{\beta}_1^a}$$

$$\Rightarrow \text{Educ}_i^b = \frac{\bar{y}^b - \hat{\beta}_0^b}{\hat{\beta}_1^b}$$

where y_i = log wage of person i , $\text{Imm}_i = 1$ if person i is an immigrant, and Educ_i = years of schooling. Models (1)-(2) are estimated on the pooled sample, including immigrants and natives. Model (3)-(4) are estimated separately on natives and immigrants, respectively. Let "a" in (3) denote natives, and "b" in (4) denote immigrants. The table below presents the sample estimates.

	a		b	
	Pooled Model: Fit to Natives and Immigrants		Model for Natives	Model for Immigrants
	(1)	(2)	(3)	(4)
Constant	3.013 $\hat{\pi}_0$ (0.006)	1.546 $\hat{\beta}_0$ (0.025)	1.365 $\hat{\beta}_0^a$ (0.033)	1.676 $\hat{\beta}_0^b$ (0.035)
Immigrant	-0.247 $\hat{\pi}_1$ (0.013)	-0.072 $\hat{\gamma}$ (0.013)	--	--
Education (yrs)	--	0.104 $\hat{\beta}_1$ (0.002)	0.117 $\hat{\beta}_1^a$ (0.002)	0.088 $\hat{\beta}_1^b$ (0.002)
MSE	0.757	0.695	0.689	0.707
Adj. R-sq	0.018	0.173	0.146	0.208
Sample Size	19,092	19,092	14,921	4,141

Notes: Fit to data for males age 30-45 in March 2012 CPS. Dependent variable is log average hourly wage. Standard errors in parentheses.

Note: question (a)-(b) use information from the regression table.

(a) (10 points) What is the average log wage gap between immigrants and natives, $\bar{y}^b - \bar{y}^a$? Provide a numeric answer and explain.

(b) (15 points) On average immigrants have 12 years of schooling ($\overline{Educ}^b = 12$), and natives have 14 years of schooling ($\overline{Educ}^a = 14$). How much can the log wage gap in (a) be explained by differences in education attainment? Provide a numeric answer and explain. *Hint: You may use either the pooled regression (2) or separate models (3)-(4).*

(c) (15 points) The pooled estimate of return to education, $\hat{\beta}_1$ in model (2) is a weighted average of $\hat{\beta}_1^a$ in model (3) and $\hat{\beta}_1^b$ in model (4):

$$\hat{\beta}_1 = \omega_a \hat{\beta}_1^a + \omega_b \hat{\beta}_1^b$$

For ease of notation, define $x_i = Educ_i$. By Frisch-Waugh Theorem for OLS in sample, we have:

$$\hat{\beta}_1 = \frac{\frac{1}{N} \sum_i y_i \hat{\xi}_i}{\frac{1}{N} \sum_i \hat{\xi}_i^2}$$

where $\hat{\xi}_i = x_i - \hat{\pi}_0 - \hat{\pi}_1 Imm_i$ is the residual from the auxiliary regression. Define:

- N^a = number of natives in sample, $\hat{p}^a = \frac{N^a}{N}$ = fraction of natives in sample
- $\hat{var}(x_i|a) = \frac{1}{N^a} \sum_{i \in a} (x_i - \bar{x}^a)^2$ is the sample variance of education among natives.
 $\hat{var}(x_i|b)$ is the sample variance among immigrants.
- $\hat{var}(\hat{\xi}_i) = \frac{1}{N} \sum_i \hat{\xi}_i^2$ is the sample variance of the residual from the auxiliary regression.

Question: Find the weights ω_a, ω_b as functions of $\hat{p}^a, \hat{var}(\hat{\xi}_i)$, and group-specific variance $\hat{var}(x_i|a)$ among natives, and $\hat{var}(x_i|b)$ among immigrants.

Hint: Find $\hat{\pi}_0, \hat{\pi}_1$ in the auxiliary regression. Plug them into the numerator of $\hat{\beta}_1$ and see how the numerator of $\hat{\beta}_1$ is related to $\hat{\beta}_1^a, \hat{\beta}_1^b$.